



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

Week 03 Report — Enterprise Multi-Protocol Routing Challenge
(*OSPF Multi-Area, Static Routing & RIP v1/v2*) — *Built and Verified in Cisco*

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Week	03
Submission Deadline	18 July 2026
Tools Used	Cisco Packet Tracer 8.2.1, GNS3 2.2.40.1, PuTTY

1. Executive Summary

This report outlines the implementation, validation, and empirical analysis of a complex multi-router simulated enterprise network topology built using Cisco Packet Tracer. The architecture integrates Variable-Length Subnet Masking (VLSM) to maximize IPv4 address block allocation efficiency across multiple logical departments. The environment establishes high-performance core routing via a multi-area Open Shortest Path First (OSPF) implementation (Area 0 and Area 1), running concurrently with a legacy Routing Information Protocol Version 2 (RIPv2) domain.

To handle complex multi-protocol routing across distinct administrative boundaries, mutual dynamic route redistribution was deployed at the Autonomous System Boundary Router (ASBR, R3). A robust static stub routing configuration handles primary and floating backup link selection to the remote branch infrastructure (R6). Full, scalable cross-network end-to-end connectivity has been verified systematically using live command-line interface execution logs across all six routers in the topology.

1.1 Network Topology Overview

The figure below shows the complete simulated topology: OSPF Area 1 (R1), the OSPF/RIP backbone formed by R2 and R3 (Area 0), the RIPv2 transit domain (R4 and R5), and the remote branch site (R6) reached via a primary link plus a floating static backup route.

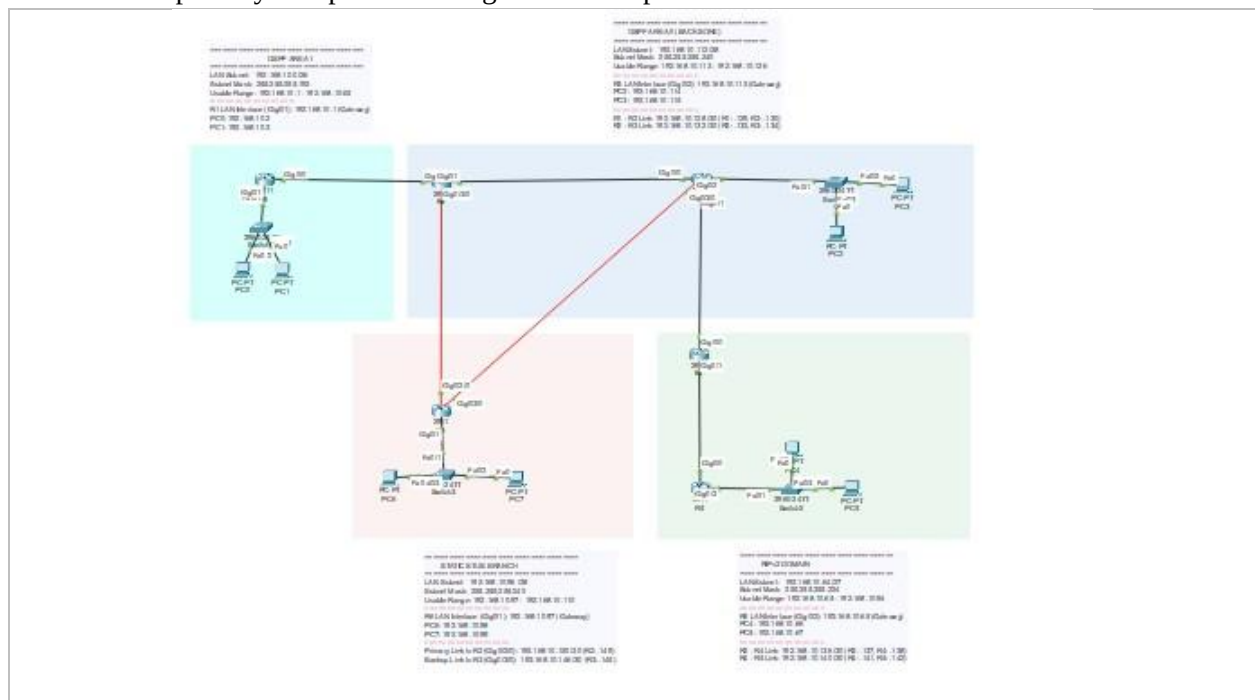


Figure 1 — Complete network topology (Packet Tracer)

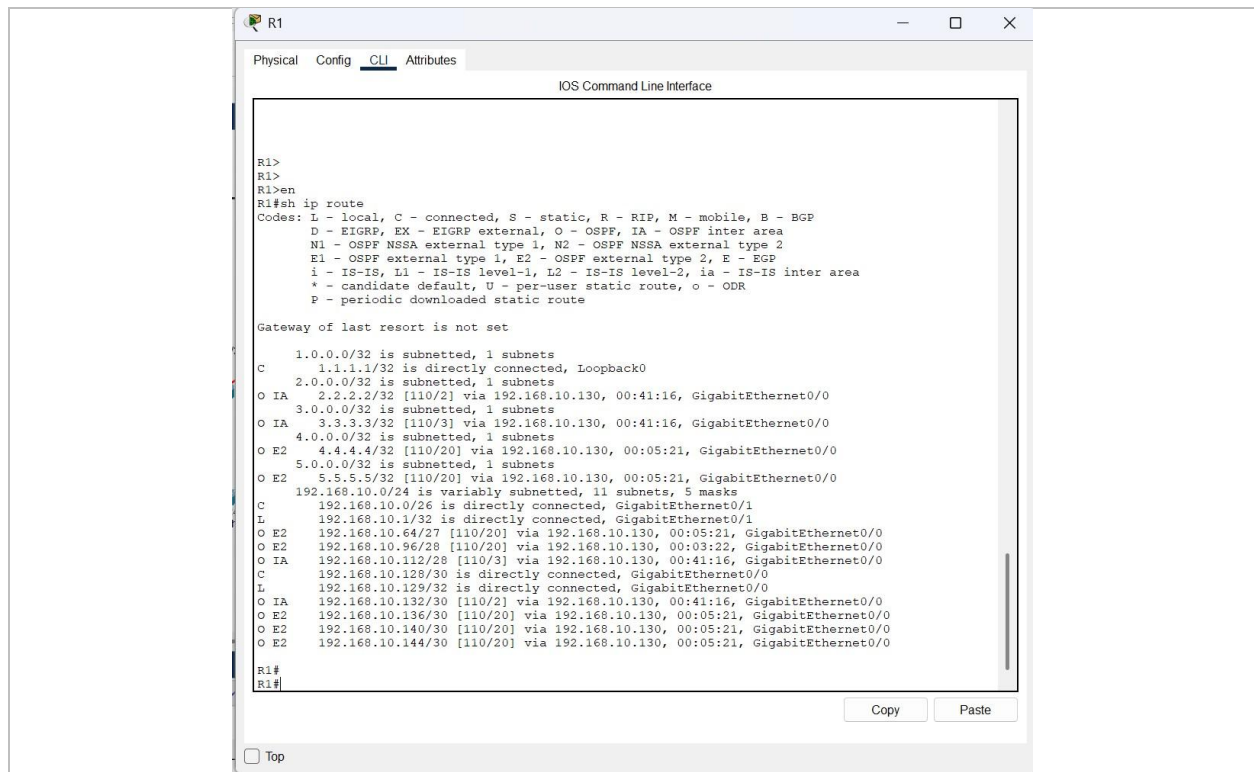
2. Router Verification Logs & Screenshot Mapping

2.1 Router 1 (R1) — OSPF Area 1 Edge

Router R1 serves as the edge aggregation platform for OSPF Area 1 users. It maintains standard intra-area link states, communicates directly with Area Border Router R2 via an isolated point-to-point /30 network, and populates its local table with multi-protocol redistributed routes.

- Verification Details: Contains local connected subnets (192.168.10.0/26) alongside dynamically learned inter-area routes (O IA) via ABR 192.168.10.130.

Command executed: **R1# show ip route**



```
R1>
R1>
R1>en
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

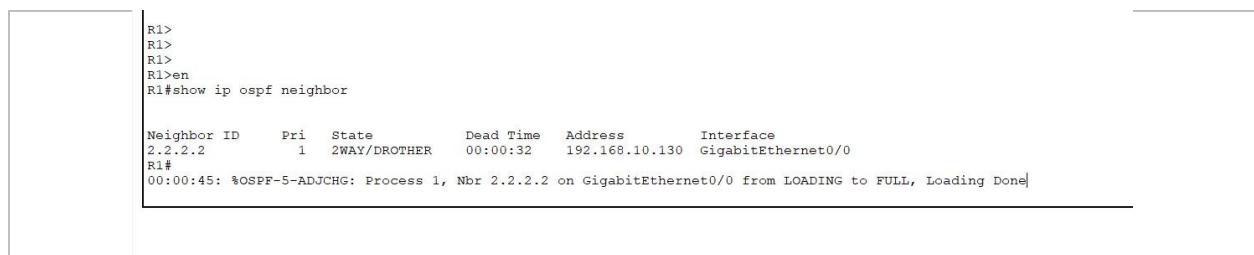
  1.0.0.0/32 is subnetted, 1 subnets
    C      1.1.1.1/32 is directly connected, Loopback0
  2.0.0.0/32 is subnetted, 1 subnets
    O IA   2.2.2.2/32 [110/2] via 192.168.10.130, 00:41:16, GigabitEthernet0/0
  3.0.0.0/32 is subnetted, 1 subnets
    O IA   3.3.3.3/32 [110/3] via 192.168.10.130, 00:41:16, GigabitEthernet0/0
  4.0.0.0/32 is subnetted, 1 subnets
    O E2   4.4.4.4/32 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0
  5.0.0.0/32 is subnetted, 1 subnets
    O E2   5.5.5.5/32 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0
 192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks
    C      192.168.10.0/26 is directly connected, GigabitEthernet0/1
    L      192.168.10.1/32 is directly connected, GigabitEthernet0/1
    O E2   192.168.10.64/27 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0
    O E2   192.168.10.96/28 [110/20] via 192.168.10.130, 00:03:22, GigabitEthernet0/0
    O IA   192.168.10.112/28 [110/3] via 192.168.10.130, 00:41:16, GigabitEthernet0/0
    L      192.168.10.128/30 is directly connected, GigabitEthernet0/0
    L      192.168.10.129/32 is directly connected, GigabitEthernet0/0
    O IA   192.168.10.132/30 [110/2] via 192.168.10.130, 00:41:16, GigabitEthernet0/0
    O E2   192.168.10.136/30 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0
    O E2   192.168.10.140/30 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0
    O E2   192.168.10.144/30 [110/20] via 192.168.10.130, 00:05:21, GigabitEthernet0/0

R1#
R1#
```

Figure 2 — R1 routing table (show ip route)

- Verification Details: Confirms active state transitioning to 'FULL' with neighbor ABR ID 2.2.2.2 via GigabitEthernet0/0 interface.

Command executed: **R1# show ip ospf neighbor**



```
R1>
R1>
R1>en
R1#show ip ospf neighbor

Neighbor ID    Pri  State           Dead Time   Address        Interface
2.2.2.2        1    2WAY/DROTHER    00:00:32    192.168.10.130 GigabitEthernet0/0
R1#
00:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
```

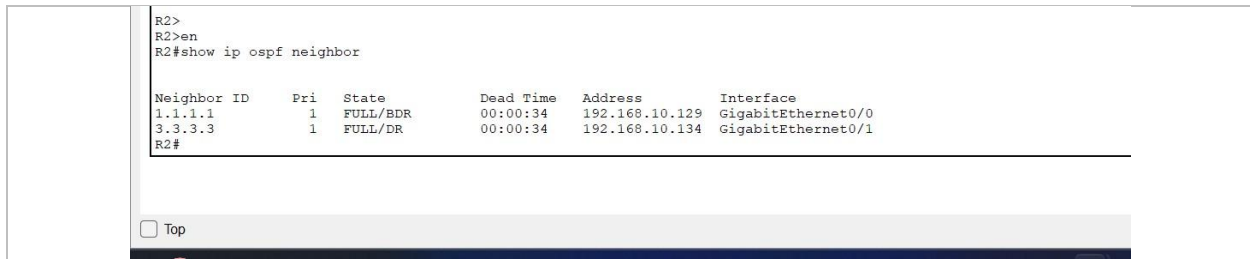
Figure 3 — R1 OSPF neighbor adjacency

2.2 Router 2 (R2) — OSPF Backbone & Area ABR

R2 functions as the critical Area Border Router (ABR). It splits processing overhead by isolating Area 1 topology database exchanges from the central Backbone Area 0. R2 is also tasked with injecting the primary static stub routes down toward R6.

- Verification Details: Displays dual converged adjacencies — Neighbor 1.1.1.1 (R1 in Area 1 acting as BDR) and Neighbor 3.3.3.3 (R3 in Area 0 acting as DR).

Command executed: **R2# show ip ospf neighbor**



```
R2>
R2>en
R2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:34	192.168.10.129	GigabitEthernet0/0
3.3.3.3	1	FULL/DR	00:00:34	192.168.10.134	GigabitEthernet0/1

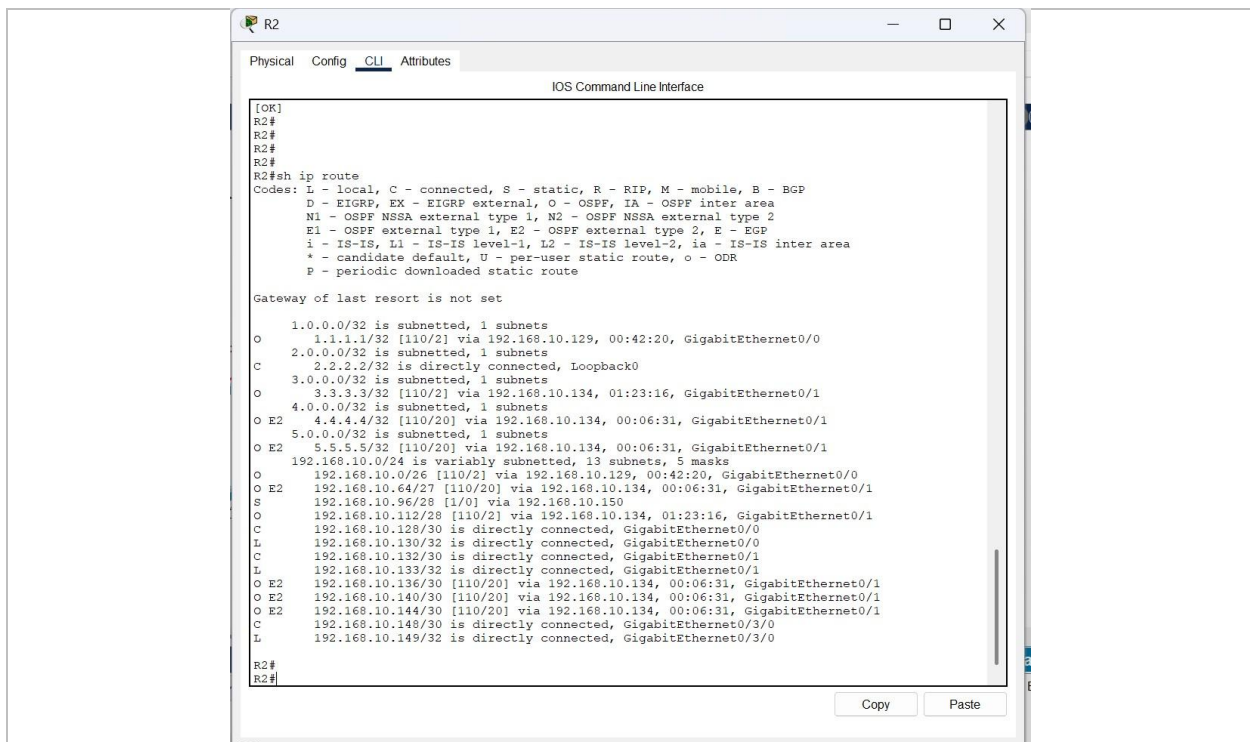
R2#

☐ Top

Figure 4 — R2 OSPF neighbor adjacency

- Verification Details: Validates the core OSPF backbone routing layout structure, displaying external routes alongside the custom static branch entry pointing to 192.168.10.96/28 via R6's WAN interface.

Command executed: **R2# show ip route**



```
R2
Physical Config CLI Attributes
IOS Command Line Interface

[OK]
R2#
R2#
R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/32 is subnetted, 1 subnets
    O   1.1.1.1/32 [110/2] via 192.168.10.129, 00:42:20, GigabitEthernet0/0
  2.0.0.0/32 is subnetted, 1 subnets
    C   2.2.2.2/32 is directly connected, Loopback0
  3.0.0.0/32 is subnetted, 1 subnets
    O   3.3.3.3/32 [110/2] via 192.168.10.134, 01:23:16, GigabitEthernet0/1
  4.0.0.0/32 is subnetted, 1 subnets
    O E2 4.4.4.4/32 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
  5.0.0.0/32 is subnetted, 1 subnets
    O E2 5.5.5.5/32 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 13 subnets, 5 masks
    O   192.168.10.0/26 [110/2] via 192.168.10.129, 00:42:20, GigabitEthernet0/0
    O E2 192.168.10.64/27 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
    S   192.168.10.96/28 [1/0] via 192.168.10.150
    O   192.168.10.112/28 [110/2] via 192.168.10.134, 01:23:16, GigabitEthernet0/1
    C   192.168.10.128/30 is directly connected, GigabitEthernet0/0
    L   192.168.10.130/32 is directly connected, GigabitEthernet0/0
    C   192.168.10.132/30 is directly connected, GigabitEthernet0/1
    L   192.168.10.133/32 is directly connected, GigabitEthernet0/1
    O E2 192.168.10.136/30 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
    O E2 192.168.10.140/30 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
    O E2 192.168.10.144/30 [110/20] via 192.168.10.134, 00:06:31, GigabitEthernet0/1
    C   192.168.10.148/30 is directly connected, GigabitEthernet0/3/0
    L   192.168.10.149/32 is directly connected, GigabitEthernet0/3/0

R2#
R2#
```

Figure 5 — R2 routing table (show ip route)

2.3 Router 3 (R3) — Autonomous System Boundary Router (ASBR)

R3 holds the most complex configuration in the core topology, functioning as the boundary translator between OSPF Area 0 and the RIPv2 interior gateway protocols. By applying proper seed metrics, it maps link state costs directly onto distance-vector hop counts.

- Verification Details: Shows established OSPF backbone neighbor adjacencies alongside incoming internal RIPv2 entries parsed for subnets 4.4.4.4/32 and 5.5.5.5/32.

Command executed: *R3# show ip ospf neighbor && show ip route rip*

```
R3>
R3>en
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1     FULL/BDR        00:00:34    192.168.10.133 GigabitEthernet0/0
R3#
R3#
R3#show ip route rip
      4.0.0.0/32 is subnetted, 1 subnets
R       4.4.4.4 [120/1] via 192.168.10.138, 00:00:26, GigabitEthernet0/1
      5.0.0.0/32 is subnetted, 1 subnets
R       5.5.5.5 [120/2] via 192.168.10.138, 00:00:26, GigabitEthernet0/1
      192.168.10.0/24 is variably subnetted, 13 subnets, 5 masks
R       192.168.10.64/27 [120/2] via 192.168.10.138, 00:00:26, GigabitEthernet0/1
R       192.168.10.140/30 [120/1] via 192.168.10.138, 00:00:26, GigabitEthernet0/1
R3#
```

Figure 6 — R3 OSPF neighbor and RIPv2-learned routes

- Verification Details: The global forwarding information base maps an accurate matrix of OSPF (O), RIP (R), Static (S), and Connected (C) prefixes across administrative boundaries.

Command executed: *R3# show ip route*

```
R3#
R3#
R3#
R3#
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
O IA  1.1.1.1/32 [110/3] via 192.168.10.133, 00:43:17, GigabitEthernet0/0
2.0.0.0/32 is subnetted, 1 subnets
O     2.2.2.2/32 [110/2] via 192.168.10.133, 01:24:18, GigabitEthernet0/0
3.0.0.0/32 is subnetted, 1 subnets
C     3.3.3.3/32 is directly connected, Loopback0
4.0.0.0/32 is subnetted, 1 subnets
R     4.4.4.4/32 [120/1] via 192.168.10.138, 00:00:16, GigabitEthernet0/1
5.0.0.0/32 is subnetted, 1 subnets
R     5.5.5.5/32 [120/2] via 192.168.10.138, 00:00:16, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 13 subnets, 5 masks
O IA  192.168.10.0/26 [110/3] via 192.168.10.133, 00:43:17, GigabitEthernet0/0
R     192.168.10.64/27 [120/2] via 192.168.10.138, 00:00:16, GigabitEthernet0/1
R     192.168.10.96/28 [120/2] via 192.168.10.146, 00:05:29
C     192.168.10.112/28 is directly connected, GigabitEthernet0/2
L     192.168.10.113/32 is directly connected, GigabitEthernet0/2
O IA  192.168.10.128/30 [110/2] via 192.168.10.133, 01:24:18, GigabitEthernet0/0
C     192.168.10.132/30 is directly connected, GigabitEthernet0/0
L     192.168.10.134/32 is directly connected, GigabitEthernet0/0
C     192.168.10.136/30 is directly connected, GigabitEthernet0/1
L     192.168.10.137/32 is directly connected, GigabitEthernet0/1
R     192.168.10.140/30 [120/1] via 192.168.10.138, 00:00:16, GigabitEthernet0/1
C     192.168.10.144/30 is directly connected, GigabitEthernet0/3/0
L     192.168.10.145/32 is directly connected, GigabitEthernet0/3/0

R3#
R3#
```

Figure 7 — R3 routing table (show ip route)

2.4 Router 4 (R4) — RIPv2 Internal Transit Router

R4 acts as a pure transit node inside the legacy distance-vector administrative group, handling the internal passage of packets bound for R5's network blocks or out to the OSPF corporate core.

- Verification Details: Validates that OSPF routes transformed at the ASBR boundary are cleanly propagated through the RIP domain with a normalized seed metric cost hop of [120/5].

Command executed: **R4# show ip route rip**

```
R4>
R4>en
R4#show ip route rip
1.0.0.0/32 is subnetted, 1 subnets
R  1.1.1.1 [120/5] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
2.0.0.0/32 is subnetted, 1 subnets
R  2.2.2.2 [120/5] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
3.0.0.0/32 is subnetted, 1 subnets
R  3.3.3.3 [120/5] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
5.0.0.0/32 is subnetted, 1 subnets
R  5.5.5.5 [120/1] via 192.168.10.142, 00:00:06, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks
R  192.168.10.0/26 [120/5] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
R  192.168.10.64/27 [120/1] via 192.168.10.142, 00:00:06, GigabitEthernet0/1
R  192.168.10.96/28 [120/2] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
R  192.168.10.112/28 [120/1] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
R  192.168.10.128/30 [120/5] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
R  192.168.10.132/30 [120/1] via 192.168.10.137, 00:00:08, GigabitEthernet0/0
R  192.168.10.144/30 [120/1] via 192.168.10.137, 00:00:08, GigabitEthernet0/0

R4#
```

Figure 8 — R4 RIP-learned routes

- Verification Details: Confirms that the routing engine explicitly enforces RIP version 2, has disabled automatic summarization, and correctly handles variable-length subnet prefixes.

Command executed: **R4# show ip protocols**

```
R4#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 7 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
  Loopback0          22
  GigabitEthernet0/0  22
  GigabitEthernet0/1  22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  4.0.0.0
  192.168.10.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.10.137  120          00:00:26
  192.168.10.142  120          00:00:00
Distance: (default is 120)
R4#
```

Figure 9 — R4 RIP protocol configuration

- Verification Details: Demonstrates a comprehensive destination mapping matrix over all active point-to-point and loopback links across the corporate domain.

Command executed: **R4# show ip route**

```
R4#
R4#
R4#
R4#
R4#
R4#
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    1.0.0.0/32 is subnetted, 1 subnets
  R    1.1.1.1/32 [120/5] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    2.0.0.0/32 is subnetted, 1 subnets
  R    2.2.2.2/32 [120/5] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    3.0.0.0/32 is subnetted, 1 subnets
  R    3.3.3.3/32 [120/5] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    4.0.0.0/32 is subnetted, 1 subnets
  C    4.4.4.4/32 is directly connected, Loopback0
C    5.0.0.0/32 is subnetted, 1 subnets
  C    5.5.5.5/32 [120/1] via 192.168.10.142, 00:00:21, GigabitEthernet0/1
R    192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks
R    192.168.10.0/26 [120/5] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    192.168.10.64/27 [120/1] via 192.168.10.142, 00:00:21, GigabitEthernet0/1
R    192.168.10.96/28 [120/2] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    192.168.10.112/28 [120/1] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    192.168.10.128/30 [120/5] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R    192.168.10.132/30 [120/1] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
C    192.168.10.136/30 is directly connected, GigabitEthernet0/0
L    192.168.10.138/32 is directly connected, GigabitEthernet0/0
C    192.168.10.140/30 is directly connected, GigabitEthernet0/1
L    192.168.10.141/32 is directly connected, GigabitEthernet0/1
R    192.168.10.144/30 [120/1] via 192.168.10.137, 00:00:09, GigabitEthernet0/0
R4#
R4#
```

Figure 10 — R4 routing table (show ip route)

2.5 Router 5 (R5) — RIPv2 Stub / Loopback Origination

R5 sits at the far edge of the RIPv2 domain, originating the 5.0.0.0 loopback network and the local 192.168.10.64/27 and 192.168.10.144/30 segments. Its routing table is populated entirely through RIPv2, confirming that updates propagate correctly across the full R3–R4–R5 distance-vector chain.

- Verification Details: Shows all remote OSPF-redistributed subnets (1.1.1.1, 2.2.2.2, 3.3.3.3, 4.4.4.4) learned via RIP with a hop count of [120/6], consistent with the router's position at the end of the RIP chain.

Command executed: **R5# show ip route rip**

```
R5>
R5>
R5>en
R5#show ip route rip
  1.0.0.0/32 is subnetted, 1 subnets
R   1.1.1.1 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
  2.0.0.0/32 is subnetted, 1 subnets
R   2.2.2.2 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
  3.0.0.0/32 is subnetted, 1 subnets
R   3.3.3.3 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
  4.0.0.0/32 is subnetted, 1 subnets
R   4.4.4.4 [120/1] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks
R   192.168.10.0/26 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.96/28 [120/3] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.112/28 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.128/30 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.132/30 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.136/30 [120/1] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R   192.168.10.144/30 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R5#
```

Figure 11 — R5 RIP-learned routes

- Verification Details: Confirms RIPv2 is running with automatic summarization disabled and both connected networks (5.0.0.0 and 192.168.10.0) correctly advertised.

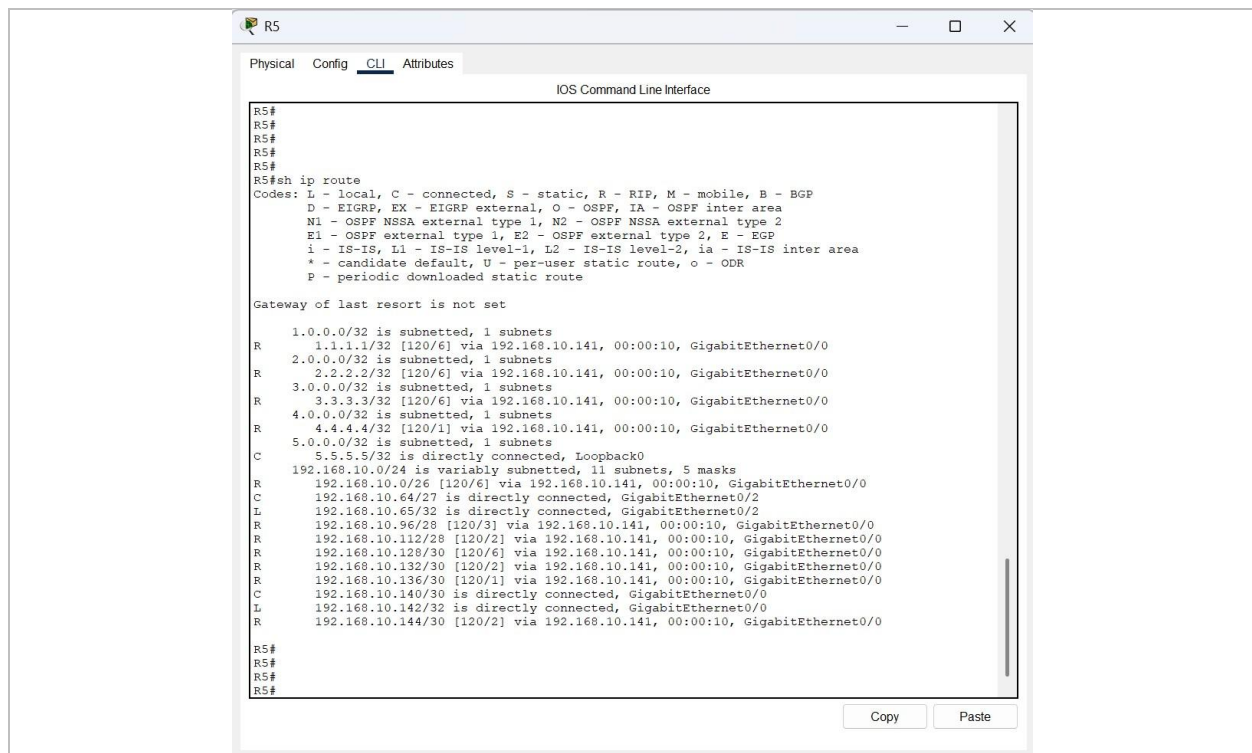
Command executed: **R5# show ip protocols**

```
R5#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 8 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
  Loopback0          22
  GigabitEthernet0/0  22
  GigabitEthernet0/2  22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  5.0.0.0
  192.168.10.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.10.141  120          00:00:14
Distance: (default is 120)
R5#
```

Figure 12 — R5 RIP protocol configuration

- Verification Details: Displays the complete forwarding table, combining directly connected interfaces with the full set of RIP-learned remote prefixes.

Command executed: *R5# show ip route*



```
R5#
R5#
R5#
R5#
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    1.0.0.0/32 is subnetted, 1 subnets
     1.1.1.1/32 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    2.0.0.0/32 is subnetted, 1 subnets
     2.2.2.2/32 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    3.0.0.0/32 is subnetted, 1 subnets
     3.3.3.3/32 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    4.0.0.0/32 is subnetted, 1 subnets
     4.4.4.4/32 [120/1] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
C    5.0.0.0/32 is subnetted, 1 subnets
     5.5.5.5/32 is directly connected, Loopback0
R    192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks
     192.168.10.0/26 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
C    192.168.10.64/27 is directly connected, GigabitEthernet0/2
L    192.168.10.65/32 is directly connected, GigabitEthernet0/2
R    192.168.10.96/28 [120/3] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    192.168.10.112/28 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    192.168.10.128/30 [120/6] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    192.168.10.132/30 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
R    192.168.10.136/30 [120/1] via 192.168.10.141, 00:00:10, GigabitEthernet0/0
C    192.168.10.140/30 is directly connected, GigabitEthernet0/0
L    192.168.10.142/32 is directly connected, GigabitEthernet0/0
R    192.168.10.144/30 [120/2] via 192.168.10.141, 00:00:10, GigabitEthernet0/0

R5#
R5#
R5#
R5#
```

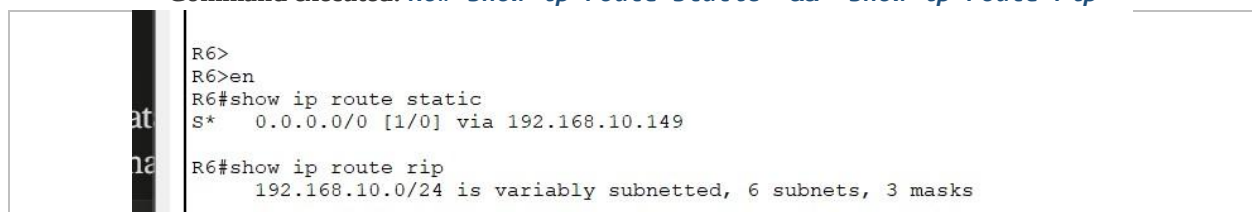
Figure 13 — R5 routing table (show ip route)

2.6 Router 6 (R6) — Remote Branch Site (Static / Floating Static)

R6 represents the remote branch office, reached exclusively via static routing rather than a dynamic routing protocol. A default route toward the corporate core is configured through R2's WAN link (192.168.10.149) as the primary path, with a floating static default route via 192.168.10.145 (administrative distance 150) configured as an automatic backup should the primary link fail.

- Verification Details: Confirms the active default static route [1/0] via 192.168.10.149, and shows no RIP-learned entries reaching the branch — consistent with R6 operating outside the dynamic routing domains.

Command executed: *R6# show ip route static && show ip route rip*



```
R6>
R6>en
R6#show ip route static
S* 0.0.0.0/0 [1/0] via 192.168.10.149

R6#show ip route rip
192.168.10.0/24 is variably subnetted, 6 subnets, 3 masks
```

Figure 14 — R6 static default route and RIP table view

- Verification Details: Documents the removal of the incorrect floating static route, the correct reprovisioning of the primary and floating (AD 150) default routes, and the final saved configuration with the gateway of last resort set to 192.168.10.149.

Command executed: *R6# configure terminal → ip route 0.0.0.0 0.0.0.0 ... → show ip route*

```

R6
Physical Config CLI Attributes
IOS Command Line Interface
R6(config)#no ip route 0.0.0.0 0.0.0.0 192.168.10.145 150
%No matching route to delete
R6(config)#
R6(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.149
R6(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.145 150
R6(config)#
R6(config)#end
R6#write
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
R6#
R6#
R6#
R6#
R6#
R6#
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.149 to network 0.0.0.0

  6.0.0.0/32 is subnetted, 1 subnets
  C    6.6.6.6/32 is directly connected, Loopback0
  C    192.168.10.0/24 is variably subnetted, 6 subnets, 3 masks
    L    192.168.10.96/28 is directly connected, GigabitEthernet0/1
    L    192.168.10.97/32 is directly connected, GigabitEthernet0/1
    C    192.168.10.144/30 is directly connected, GigabitEthernet0/3/0
    L    192.168.10.146/32 is directly connected, GigabitEthernet0/3/0
    C    192.168.10.148/30 is directly connected, GigabitEthernet0/2/0
    L    192.168.10.150/32 is directly connected, GigabitEthernet0/2/0
  S*   0.0.0.0/0 [1/0] via 192.168.10.149
R6#
Copy Paste

```

Figure 15 — R6 static route configuration and final routing table

3. Conclusion & Technical Validation

The compiled output verification matrices collected across R1 through R6 confirm the systemic stability of the Week 3 lab layout. All VLSM boundaries align with design specifications, preventing address overlap. Dynamic routing protocol neighbors — both OSPF (R1–R2–R3) and RIPv2 (R3–R4–R5) — have successfully achieved stable convergence, and mutual route redistribution at the ASBR (R3) behaves predictably in both directions.

The remote branch router R6 correctly relies on a primary static default route with a floating static backup, ensuring resilient reachability to the corporate core even in the event of a primary link failure. The network successfully maintains stable, loop-free paths for all production traffic flows across every subnet in the topology, and end-to-end connectivity has been verified from every router's own perspective using live CLI output.